

GeckoRov

Subsurface Exploration Robot

Team 08

Principal Investigator Subject Matter Expert

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Core Team Members:

Diego Bermejo, Shriya Gattapally, Connor McCullum-Zempel, Tim Badum, Joshua Medel,
Abdelrahman Helal, Arturo Flores, Michael Gamota, Pranav Ananthram

NASA 2020 Taxonomy Identifiers:

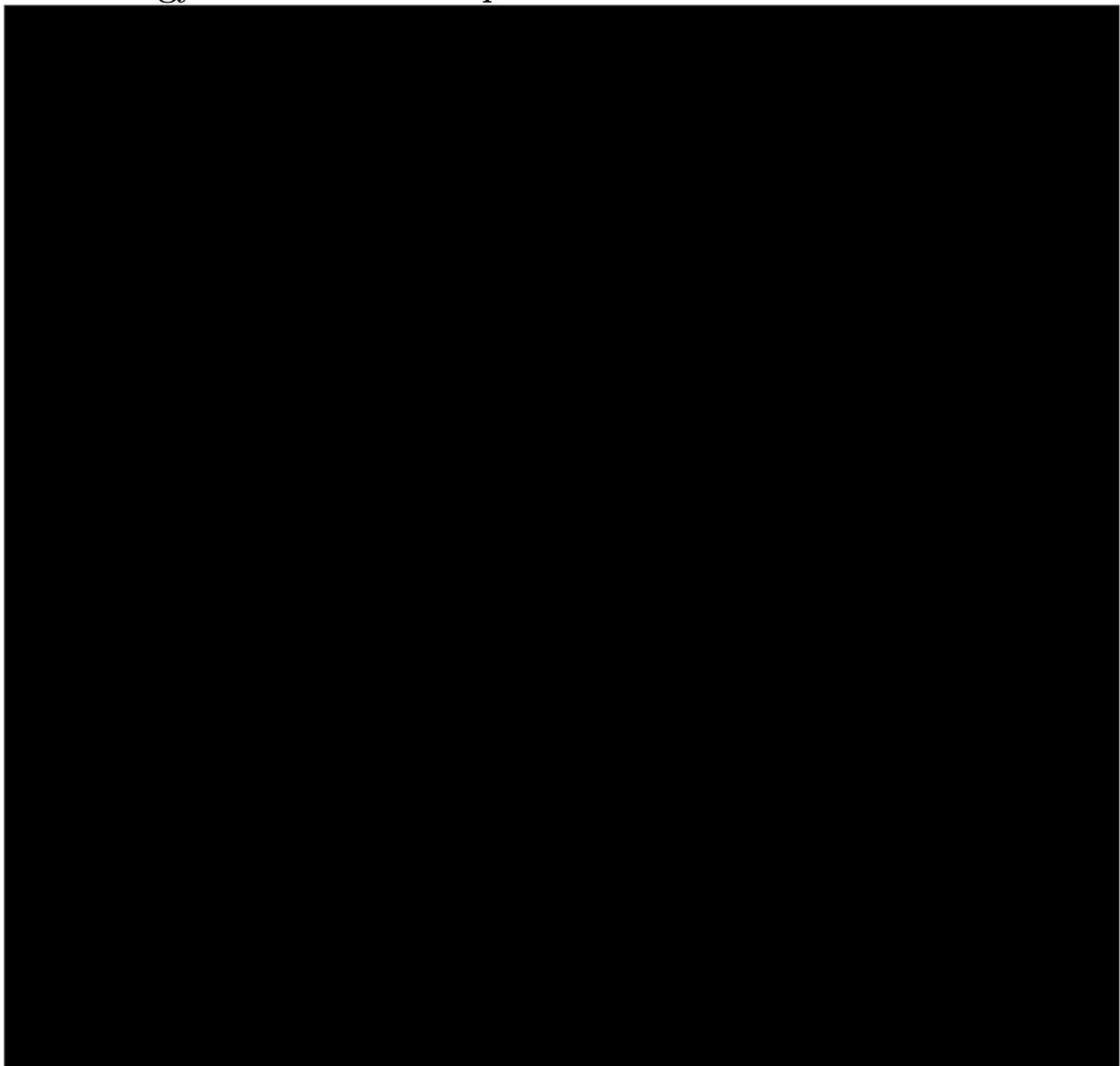
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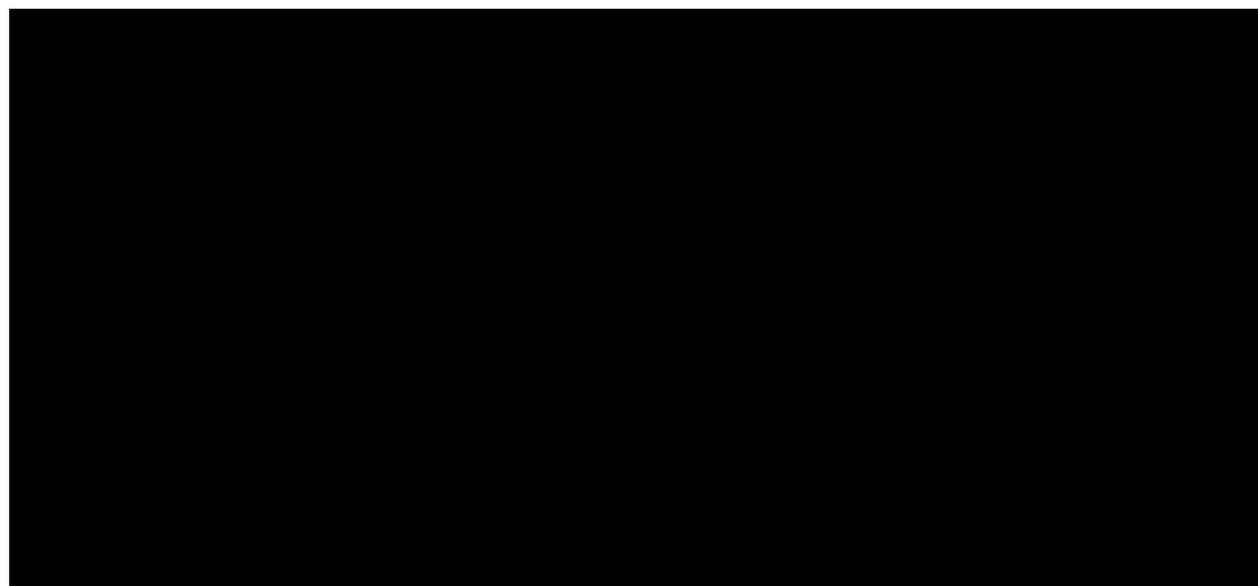
July 19, 2023

Abstract

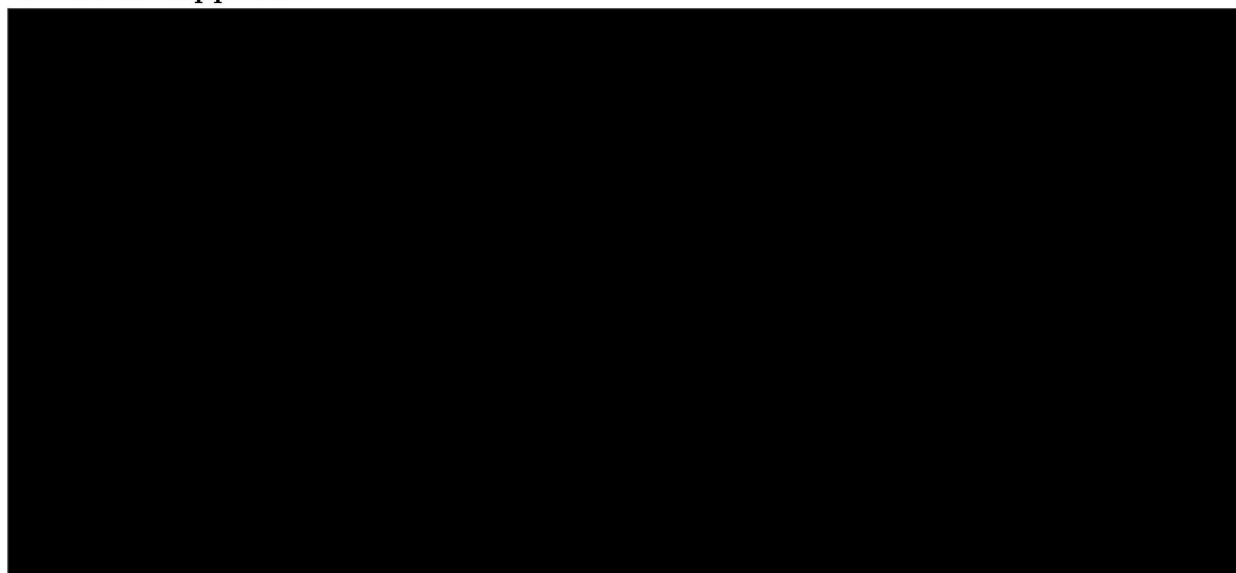
NASA's Mars Program has been advancing for years, and the robots sent to Mars played a major role in this program and gave us a deeper understanding of the planet's surface, environment, and chemical structure. Nevertheless, we are looking beyond the surface for high terrains, caves, and subsurface structures. Accordingly, the robots need to be able to climb hills and terrains and be small enough to explore caves and go beneath the surface. Our proposal aims to research the combination of existing wall climbing robots and self-folding soft robots. The research includes the materials and designs that will be appropriate for both mechanisms in addition to their effectiveness compared to existing Mars exploration robots. Specifically, we will be testing the integration between Shape Memory Polymers and micro-grippers to design a Mars robot with less than 10 lbs and less than one fifth the volume of existing rovers like ingenuity. Therefore, it will be easier to launch and operate the robot that will have high flexibility in exploring Mars surface and subsurface.

Technology Merit and Workplan

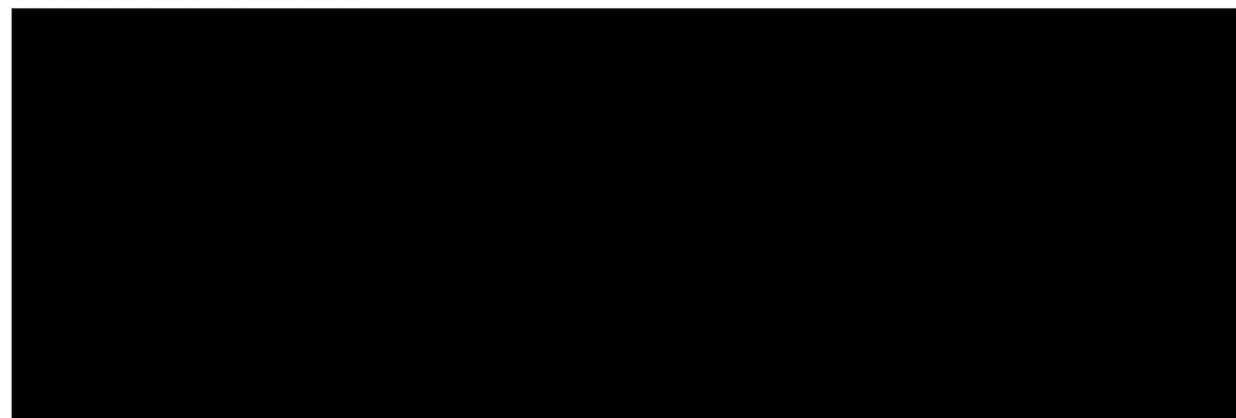


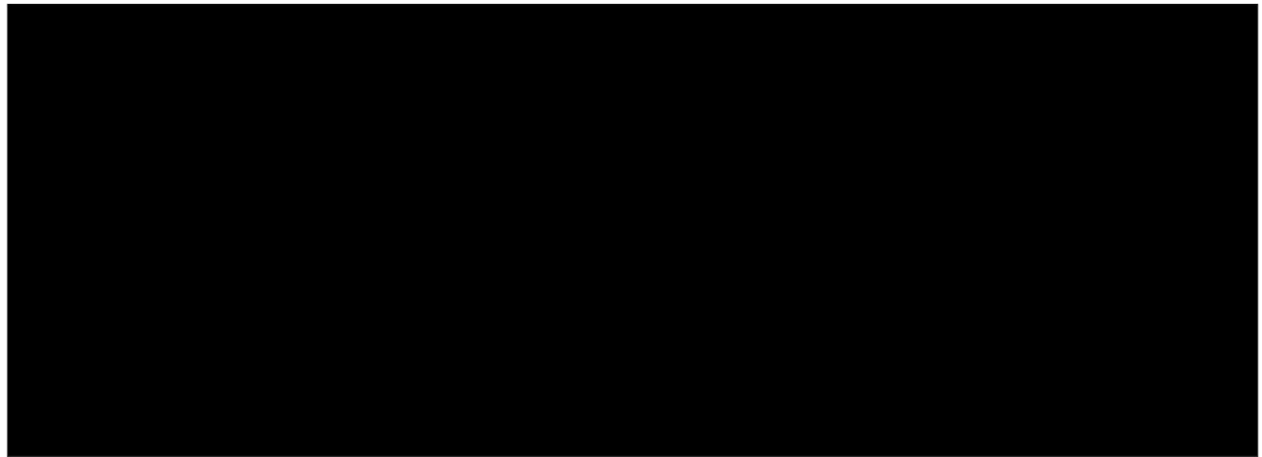


Technical Approach



Benefits and Outcomes



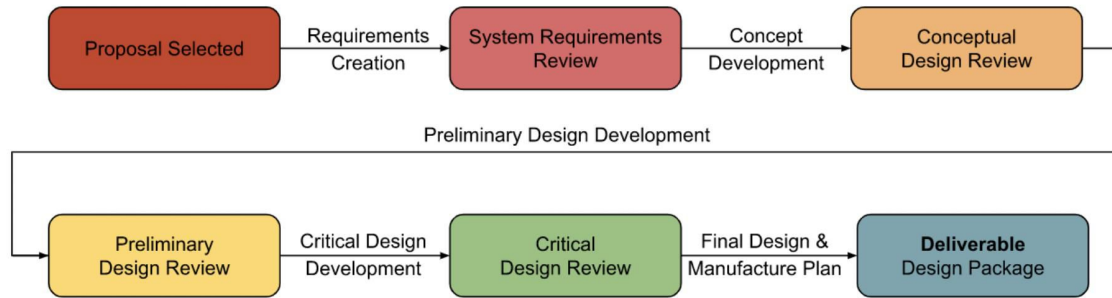


Project Management Approach

Role	Responsibilities
Principal Investigator	Shall approve technical progress through regular design reviews. Shall further outreach to prospective technology partners. Shall set system-level objectives
Project Manager	Shall implement PI system-level objectives through the management of the entire team. Shall assign tasks to leads for further implementation
Lead Scientist	Shall further research through management of scientists and translating system-level objectives to research quotas
Lead Engineer	Shall organize system-level objectives, and assign tasks to engineers according to system organization
Lead Critic	Shall ensure the quality of design, through management of critics and defining system-level objectives
Business Lead	Shall assist PI in outreach, and conduct market research for design product
Physical Scientist	Shall study the physical properties of the technology, and provide physics research for engineering development
Computer Scientist	Shall write the robotic control for the engineered design, and manage all other coding related assignments
Mathematician	Shall perform mathematical analysis as needed by physical science and engineering teams
Mechanical Engineer I	Shall collaborate with the engineering team to complete mechanical design tasks given by lead engineer, while utilizing proper engineering standards throughout design
Mechanical Engineer II	Shall collaborate with the engineering team to complete mechanical design tasks given by lead engineer, while utilizing proper engineering standards throughout design
Electrical Engineer	Shall collaborate with the engineering team and computer scientist to design electronic system and integrate controls
Critic	Shall review the engineering team's design and check for quality standards as set by the NSPE's Code of Ethics

The above table illustrates the responsibilities of each role within the team. Although those who fulfill the roles may change, this responsibility chart will act as a guideline during product development.

Task	Hours	Rationale
Requirements Creation	25	Using model-based systems is necessary to keep the project on track during development. It will take 15 hours to create the proper architectures, 5 hours to define all requirements, and 5 hours to apply requirements to architectures. We will also order our materials as soon as the requirements are set.
System Requirements Review	12	Prior to the review a 10-20 slide presentation will need to be created, with a maximum of 30 minutes per slide, will take up to 10 hours, the review itself should take no longer than 2 hours
Concept Creation/Selection	20	A total of 5 concepts will be created, it is estimated that each concept will take 3 hours. A trade study will be created to select the best concept, which will take up to 5 hours.
Conceptual Design Review	22	Prior to the review a 10-20 slide presentation will need to be created, with a maximum of 1 hour per slide, will take up to 20 hours, the review itself should take no longer than 2 hours.
Robotics Structure Design	165	A CAD model consisting of 25-75 components, with a maximum of 2 hours to complete each component, and which 1 out of 10 need to be re-CADed will take up to 165 hours.
Electronic System Design	70	An electronic diagram of around 10-30 components, with a maximum of 30 minutes per component, with 1 out of 3 needed to be re-CADed, will take up to 20 hours. Selection of components will take a maximum of 2 hours per component, thus will take up to 50 hours.
Software System Design	60	Developing coding which is compatible with each component, with a maximum of 1 hour per component, will take up to 30 hours. Integrating all component software into one master executable, with a maximum of 1 hour per component will take up to 30 hours.
Preliminary Design Review	32	Prior to the review a 20-30 slide presentation will need to be created, with a maximum of 1 hour per slide, will take up to 30 hours, the review itself should take no longer than 2 hours.
Detailed Mechanical Design	165	CAD revision, assuming 1 out of 10 components need to be re-CADed, with a maximum of 2 hours per component, will take up to 15 hours. Structural and thermal analysis of existing CAD, with a maximum of 1 hour per component, will take up to 75 hours. Fastener selection and interface design of CAD model, with a maximum of 1 hour per component, will take up to 75 hours.
Power Analysis	30	Power analysis of each electronic component, with 10-30 components, at a maximum of 1 hour per component, will take up to 30 hours.
Control Systems Design	105	Basic control modeling of 5-10 controllable devices, with a maximum of 8 hours per device, will take up to 80 hours. Navigation control of the robot, with 3-5 different modes of movement, with a maximum of 5 hours per mode, will take up to 25 hours.
Critical Design Review	52	Prior to the review a 30-50 slide presentation will need to be created, with a maximum of 1 hour per slide, will take up to 50 hours, the review itself should take no longer than 2 hours.
Redesign Margin	36	CAD revision, assuming 1 out of 10 components need to be re-CADed, with a maximum of 2 hours per component, will take up to 15 hours. If a coding error occurs 1 out of 10 times, and each coding error takes 2 hours to troubleshoot, then it could take up to 21 hours.
Manufacture Plan	38	Manufacture planning, with a maximum of 30 minutes per component will take up to 38 hours.



Once selected, reviews will serve as design milestones to guide development toward a complete design package. Professionals from NASA L'Space serve as our primary stakeholders and will attend such reviews to maintain the proper direction of product development. This process is outlined in the task table above.

Cost Element	Labor Hours
Mechanical Engineering	383
Software Engineering	186
Electrical Engineering	100
Quality Control	86
Subtotal	755
Project Management	121
Total	876

Total work hours	876 hours
Work distributed amongst 10 students	x 0.10 total hours
Working part-time 10 hours / week	x 0.10 = # weeks
Holidays, setbacks, etc.	x 1.5
Weeks until deliverable	13.14

In the table above on the left, a total of 876 labor hours are calculated in order to deliver a complete design package of the TerraXplor. Program Management hours were calculated through the detailed hourly table, and through the assumption that for every 10 hours of work, 1 hour of management is needed. In the table above on the right, the time it will take our team of 10 students to deliver the design package of TerraXplor is calculated to be 13.14 weeks accounting for part-time work, holidays, and setbacks.

Teaming and Workforce Development

Personnel

As calculated prior, our deliverable is projected to take 13 weeks. Our members shall work no more than 10 hours per week, due to the part-time nature of college students. Furthermore, we want to account for mistakes, other commitments, and sudden events, thus we assume only $\frac{2}{3}$ of allotted student time will be effective. This means that no member shall be assigned to work more than 88 hours over the course of 13 weeks, this maximum time commitment was used to properly manage student time below.

**Members assigned to multiple roles are color coded*

Role	Fulfilled by	Skills/Experience	Hours
Mechanical Engineering			Total Hours: 383
Mechanical Engineer I	Arturo Flores	Mechatronics engineer, CAD, Systems & Controls	88
Mechanical Engineer II	William Harris	CAD, manufacturing	40
Lead Scientist	Shriya Gattapally	CAD	51
Physical Scientist	Joshua Medel	Experimental measurements and data analysis, python programming for data processing and analysis, CAD	88
Lead Engineer	Connor McCullum-Zempel	CAD, manufacturing, electronics	76
Lead Critic	Tim Badum	CAD, manufacturing	40
Software Engineering			Total Hours: 186
Computer Scientist	Abdelrahman Helal	Statistical data analysis and machine learning	88
Mathematician	Diego Bermejo	Functional analysis, linear programming, systems optimization, Arduino programming	30
Lead Scientist	Shriya Gattapally	CAD, Python	68
Electrical Engineering			Total Hours: 100
Electrical Engineer	Michael Gamota	Embedded systems, wireless communication, hardware design	44
Electrical Engineer	Pranav Ananthram	Electronics, Embedded systems, PCBs	44
Lead Engineer	Connor McCullum-Zempel	Instrumentation Lead of a competitive robotics team	12
Quality Control			Total Hours: 86
Critic	William Harris	PI of Project	43
Lead Critic	Tim Badum	Systems Engineer for a project team	43
Program Management			Total Hours: 121
Principal Investigator	William Harris	Co President of a competitive robotics team	45+
Project Manager	Diego Bermejo	Co-Head of competition math team	36
Business Lead	Michael Gamota	Business Minor	10
Lead Scientist	Shriya Gattapally	Led a team designing a Satellite in CAD	10
Lead Engineer	Connor McCullum-Zempel	Instrumentation Lead of a competitive robotics team	10
Lead Critic	Tim Badum	Systems Engineer for a project team	10

Facilities & Equipment

To obtain materials we would like to test, the team must allocate money to purchase and shipping. The table below shows the cost of materials according to manufacturers and online distributors.

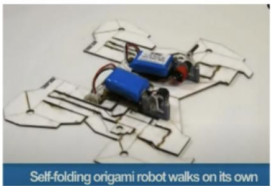
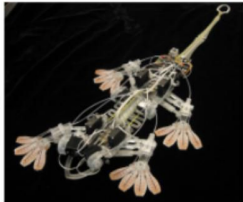
Material	Cost	Quantity	Net Cost
Thermoplastic urethan	\$5.5/ kg	2	\$11
Shape memory polymers	\$11/lb	10	\$110
Carbon fiber reinforced polymers	\$70/lb	2	\$140
Kapton film	\$25/line	5	\$125
Mylar	\$ 50/(4ft x 100ft)	1	\$50
Total			\$436

In order to deliver a professional level design package, the team needs to have access to CAD modeling software, coding software (preferably compatible with Control Systems), and a document management service.

Item	Cost	Student Modifiers	Net Cost
CAD Software	\$ 2,800.00/year	-100%	\$0.00
MATLAB Coding Software	\$ 940.00/year	-100%	\$0.00
Microsof Office Suit	\$ 439.99/year	-100%	\$0.00
Materials	\$ 436	x2.29(margin)	\$ 1000
Student Work Stipend	\$ 1000/per person	10 students	\$10000.00

Appendix

Quad Chart

Gecko & Origami Inspired Subsurface Exploration Robot PI: William Harris, Team #8 	
Goal / Objective Mars exploration is carried out by two large rovers and a drone. We cannot yet explore the subsurface of Mars beyond the imaging of the RIMFAX on Perseverance. Furthermore, the current rovers are only able to penetrate the surface up to 2.8 inches deep, which does not allow for the collection of rock samples below the sterilizing effects of surface radiation. By combining two soft robotics technologies, it is possible to explore deep into the subsurface of Mars including caves or even drilled holes as NASA improves rover capabilities. These technologies include Wall Climbing Robots (WCRs) and shape-memory composite self-folding Robots. This concept qualifies as an NTR. What we would like to produce for the \$10k, if funded, is a prototype for a wall climbing robot that can fold to fit through small cracks and crevices for primarily exploration missions on Mars. It could also potentially be used to gather rock samples from the subsurface.	• https://spectrum.ieee.org/self-folding-printable-origami-robot (left image) • https://ieeexplore.ieee.org/document/8665184 (right image)  +  Self-Folding Origami Robot Stickybot
Team Overview • Team – William Harris, Diego Bermejo, Pranav Ananthram, Colin Ro, Arturo Flores, Tim Badum, Abdelrahman Helal, Joshua Medel, Namita Shah, Connor McCullum-Zempel, Michael Gamota, Shriya Gattapally • Majors: – Mechanical Engineering, Electrical Engineering, Physics, Computer Science, Applied Mathematics, Astrophysics • Subject Matter Experts – Michael Tolley - Mechanical Engineering Professor & Soft Robotics Researcher – Kalind Carpenter - Robotics Engineer at NASA JPL • Taxonomy – TX04.2.1 – TX04.2.42	Metrics and Key Performance Parameters The future state of this concept can help improve our surface missions of Mars. Due to the low operating costs associated with creating an origami soft-robot, the technology is practical for investing money in research and deploying on long-term missions. Assuming the future state of this robot possesses the capability to squeeze through small cracks and crevices either natural or drilled, we can extend the range of missions to explore and collect samples previously unreachable to the rovers. Mars Rovers are very large, with dimensions of 10x9x7 ft³ and are only able to traverse the surface of Mars. To traverse the subsurface, we aim to make a smaller robot. Stickybot is 2x8x24 in³ and we would like to make a similar robot collapsing to a height of less than 1 in. Current folding robots are capable of folding to have a height of a few millimeters, however, the need to keep it mobile will limit its ability to collapse.

NPWEE Spring 2023

New Technology Report (NTR)
 Appended to pdf

Team Bios

William Harris

William is a third-year undergraduate mechanical engineering student and Co President of the Triton Robotics team at UC San Diego. Previously, he was also a subteam lead, in charge of one of the team's robots, which allowed him to gain experience managing teams. Through coursework and his extracurriculars, he is experienced in CAD and manufacturing.

Diego Bermejo

Diego is a third-year undergraduate mathematics major with coursework in real and complex functional analysis, Calculus in $\mathbb{R}^n$, Proof-based Linear Algebra, and applied linear programming. He has research experience in the field of tomography as it relates to medical imaging technology and the Radon Transform. He is also bi-lingual and pursuing a Spanish Minor.

Shriya Gattapally

Shriya Gattapally is a second-year undergraduate astrophysics major and applied mathematics minor. She's led a team designing a satellite in CAD software, as well as completed various other projects and using 3D printing. Furthermore, by her second year she's completed calculus, vector calculus, and linear algebra, as well as all basic physics classes. She's also experienced with python programming.

Connor McCullum-Zempel

Connor the Instrumentation lead for the Husky Robotics team, which builds Mars rovers to take to a competition for a simulated Mars environment. Because of this, he has much experience designing parts to work in hostile environments in outer space. It also has given him experience leading engineers which is valuable as the engineering team lead. He's also learned lots about manufacturing parts, CAD, electronics, and much more.

Tim Badum

Tim Badum is a mechanical engineering student specializing in system engineering. He has experience with CAD software as well as program management. He also has experience writing proposals, and is the systems engineer for his school run team.

Joshua Medel

Joshua Medel, a third year physics student, has practical experience in experimental measurements along with python programming for data processing and analysis. Furthermore, he has hands-on experience with CAD, which enables him to contribute more to the project when needed.

Abdelrahman Helal

Abdelrahman is a second-year physics and computer science student whose curriculum focuses on data analysis and statistical inference. He also has experience with modern physics research and experience in machine learning that will help programming the robot.

Pranav Ananthram

Pranav is a current rising junior and pursuing an undergraduate degree in electrical engineering with a minor in computer science, specifically, with an interest in machine learning. He has previously worked on projects such as the UW human powered submarine as part of the electrical engineering team and has hardware experience from helping construct and assemble machines as part of robotics clubs. He has worked as a peer facilitator for courses such as physics and mathematics. Through coursework, he has acquired advanced skills in JAVA, Python, CAD, RStudio, MATLAB and LTSpice.

Arturo Flores

Arturo is a mechatronics engineer specializing in Robotics Design, ROS, Machine Learning, and Control. Currently pursuing a Master's degree in Mechanical Engineering (Systems & Control) at UCLA, under Dr. Dennis Hong's supervision. His Bachelor's Degree is in Mechatronics Engineering from the Universidad Nacional de Ingenieria.

Michael Gamota

Michael is an electrical engineering student with minors in business and computer science. His experiences lie in the fields of embedded systems, wireless communication, and hardware design with applications in both traditional and hybrid electronics. He also has entrepreneurial experience in the music industry in which he was responsible for sales, marketing, and customer relations.

References

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